

PROJECT BRIEF & BUSINESS SCENARIO

Software Release Readiness System

Company: NextWave Software

Department: Development / QA / DevOps

Project Type: Multi-Agent AI System

Business Scenario

NextWave Software develops and maintains a CRM platform used by different business teams.

Before a new software version is released to production, the Development, QA, and DevOps teams need to review a large amount of information.

This includes completed features, bug fixes, QA test results, unresolved defects, known limitations, available workarounds, and release risks.

At present, technical leads manually collect and review this information before deciding whether a release should proceed.

The challenge is that release decisions should not be based only on how many tests passed.

A single serious defect, failed security test, unresolved regression, or important business risk may affect the entire release decision.

Your task is to improve this process using a Multi-Agent AI System.

Your Challenge

Build a Multi-Agent System that reviews software release information and prepares a release-readiness assessment for the Technical Lead.

Your system should use the available release information, testing evidence, defect information, and company approval rules before reaching a recommendation.

You are responsible for deciding:

- how many agents are required;
- what each agent should be responsible for;
- how information should move between agents;
- how release changes should be reviewed;
- how QA results should be interpreted;
- how open defects should affect the release decision;
- how risk should be evaluated;
- what information should appear in the final release report; and
- when human review is required.

The AI system may recommend whether a release appears Ready, Conditional, or Not Ready, but the final production release decision must remain with the Technical Lead.

BUSINESS RESOURCES & DATA SET

Resource 1: Release Overview

Use the following information as the main context for the release.

Product	NextWave CRM
Release Version	Version 2.5
Planned Release Date	18-Sep-2026
Release Type	Feature and Maintenance Release
Release Owner	Development Team
QA Owner	QA Team
Deployment Owner	DevOps Team
Target Environment	Production
Previous Version	Version 2.4
Release Status	Under Review
Main Goal	Introduce new customer management features, improve reporting, and resolve existing bugs

Resource 2: Change and Feature Dataset

The following changes are included in Version 2.5.

CH001

Type	Feature
Area	Customer Management
Title	Customer Tagging
Description	Users can create and assign custom tags to customer profiles.
Status	Completed

CH002

Type	Feature
Area	Dashboard
Title	Sales Dashboard Filter
Description	Users can filter dashboard results by date, salesperson, and region.
Status	Completed

CH003

Type	Feature
Area	Notifications
Title	Follow-up Reminder
Description	Sales staff can receive reminders for customer follow-up activities.
Status	Completed

CH004

Type	Feature
Area	Reports
Title	PDF Report Export
Description	Users can export selected reports as PDF files.
Status	Completed

CH005

Type	Feature
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Area	User Management
Title	Role Permission Update
Description	Administrators can assign more detailed permissions to user roles.
Status	Completed

CH006

Type	Bug Fix
Area	Login
Title	Session Timeout Fix
Description	Fixed an issue where some users were logged out earlier than expected.
Status	Completed

CH007

Type	Bug Fix
Area	Customer Search
Title	Search Result Fix
Description	Fixed missing results when searching customers using partial names.
Status	Completed

CH008

Type	Bug Fix
Area	Dashboard

Title	Total Calculation Fix
Description	Corrected incorrect totals displayed in monthly reports.
Status	Completed

CH009

Type	Bug Fix
Area	Email
Title	Email Formatting Fix
Description	Fixed formatting problems in automated customer emails.
Status	Completed

CH010

Type	Bug Fix
Area	Mobile
Title	Mobile Menu Fix
Description	Fixed navigation menu problems on smaller mobile screens.
Status	Completed

CH011

Type	Bug Fix
Area	Reports
Title	Date Filter Fix

Description	Corrected report date filtering across different time zones.
Status	Completed

CH012

Type	Bug Fix
Area	Notifications
Title	Duplicate Reminder Fix
Description	Prevented duplicate follow-up notifications from being generated.
Status	Completed

Resource 3: QA Test Results

Use the following test results as evidence when assessing the release.

T001

Area	Customer Management
Test Scenario	Create and assign customer tags
Result	Passed
Priority	High
Notes	Works as expected

T002

Area	Customer Management
Test Scenario	Remove customer tags
Result	Passed

Priority	Medium
Notes	No issue found

T003

Area	Dashboard
Test Scenario	Filter dashboard by date
Result	Passed
Priority	High
Notes	Results displayed correctly

T004

Area	Dashboard
Test Scenario	Filter dashboard by salesperson
Result	Passed
Priority	High
Notes	Works correctly

T005

Area	Dashboard
Test Scenario	Filter dashboard by region
Result	Passed
Priority	Medium
Notes	Works correctly

T006

Area	Notifications
Test Scenario	Generate customer follow-up reminder
Result	Passed
Priority	High
Notes	Reminder created successfully

T007

Area	Notifications
Test Scenario	Prevent duplicate reminders
Result	Passed
Priority	High
Notes	Duplicate issue resolved

T008

Area	Reports
Test Scenario	Export report as PDF
Result	Passed
Priority	High
Notes	PDF generated successfully

T009

Area	Reports
Test Scenario	Export large report as PDF

Result	Failed
Priority	Medium
Notes	Export fails when report contains more than 1,000 records

T010

Area	User Management
Test Scenario	Assign new user permissions
Result	Passed
Priority	High
Notes	Permissions applied correctly

T011

Area	Login
Test Scenario	Verify normal session timeout
Result	Passed
Priority	High
Notes	Session remains active correctly

T012

Area	Customer Search
Test Scenario	Search using partial customer name
Result	Passed
Priority	Medium
Notes	Expected results returned

T013

Area	Dashboard
Test Scenario	Verify monthly totals
Result	Passed
Priority	High
Notes	Totals correct

T014

Area	Email
Test Scenario	Verify automated email formatting
Result	Passed
Priority	Medium
Notes	Formatting issue resolved

T015

Area	Mobile
Test Scenario	Test navigation on mobile screen
Result	Passed
Priority	Medium
Notes	Navigation works correctly

T016

Area	Reports
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Test Scenario	Verify report date filtering
Result	Passed
Priority	High
Notes	Correct across tested time zones

T017

Area	Regression
Test Scenario	Existing customer creation workflow
Result	Passed
Priority	High
Notes	No regression found

T018

Area	Regression
Test Scenario	Existing sales opportunity workflow
Result	Passed
Priority	High
Notes	No regression found

T019

Area	Regression
Test Scenario	Existing reporting workflow
Result	Passed
Priority	High

Notes	No major regression found
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T020

Area	Security
Test Scenario	Verify role-based access
Result	Passed
Priority	High
Notes	Users cannot access unauthorized areas

Resource 4: Open Defects and Workarounds

The following defects are still unresolved.

BUG201

Area	Reports
Severity	Medium
Description	Large PDF exports fail when reports contain more than 1,000 records.
Business Impact	Users with very large reports cannot export them directly.
Workaround Available	Yes
Status	Open

BUG202

Area	Notifications
Severity	Medium
Description	Follow-up reminder may appear several minutes late during high system activity.

Business Impact	Some reminders may not appear immediately.
Workaround Available	Yes
Status	Open

BUG203

Area	Mobile
Severity	Low
Description	Long customer names may wrap incorrectly on small mobile screens.
Business Impact	Minor visual issue only.
Workaround Available	No
Status	Open

Available Workarounds

Defect ID	Workaround
BUG201	Users can filter the report into smaller date ranges before exporting.
BUG202	Users can refresh the notification page if a reminder does not appear immediately.
BUG203	No workaround is required because the issue does not prevent use of the system.

Resource 5: Release Risk and Approval Rules

Release Recommendation Levels

Recommendation	Meaning
Ready	No major issue prevents release and remaining risks are acceptable

Recommendation	Meaning
Conditional	Release may proceed only after specific actions, approvals, or risk acceptance
Not Ready	Important issues must be resolved before release

Defect Severity Rules

Defect Severity	Release Guidance
Critical	Release should not proceed until the defect is resolved
High	Release normally should not proceed unless there is formal approval and a strong business reason
Medium	Release may proceed if business impact is understood, a workaround exists where needed, and the risk is accepted
Low	Release may normally proceed if the issue is documented

QA Rules

Situation	Guidance
Critical test fails	Release is Not Ready
High-priority test fails	Release requires investigation before approval
Medium-priority test fails	Assess business impact and available workaround
Low-priority test fails	Document the issue and assess whether it can be accepted
Regression test fails	Review carefully because existing functionality may be affected
Security test fails	Release should not proceed until the issue is reviewed and resolved

Release Approval Rules

Your system must follow these rules.

- The AI system may assess release readiness but cannot approve a production release.
- The Technical Lead must make the final release decision.
- Any unresolved Critical or High-severity defect must be clearly highlighted.
- Medium-severity defects must include their business impact and any available workaround.
- Known issues accepted for release must be included in the final release report.
- The system must not hide or minimize failed tests simply because most tests passed.
- If release information is incomplete, conflicting, or uncertain, the system should clearly identify the uncertainty and request human review.

ASSESSMENT REQUIREMENTS & SUBMISSION

Multi-Agent Requirement

- This project must be completed as a Multi-Agent System.
- Your agents should have different and meaningful responsibilities.
- You are responsible for deciding how the work should be divided and how the agents should collaborate.
- A strong design should show that different parts of the release-readiness problem are being handled separately and then brought together in a useful final assessment.
- Simply sending the same release information through several AI nodes will not be considered a strong Multi-Agent design.

What Your System Must Demonstrate

Your completed system should demonstrate that it can:

- understand what is included in a software release;
- use QA test results as evidence;
- consider the priority of failed tests;
- connect known defects with relevant testing information;
- consider defect severity and business impact;
- recognize available workarounds;
- identify important release risks;
- apply company release rules consistently;
- avoid making a decision based only on the number of passed tests;
- identify missing or conflicting information;
- prepare a clear release-readiness assessment; and
- keep the Technical Lead in control of the final decision.

Important Rules

- Your system should use only the provided release information when forming its assessment.
- It should not invent test results.
- It should not invent defects.
- It should not assume that an unresolved defect is acceptable simply because a workaround exists.
- It should not assume that all failed tests have the same level of importance.
- It should not automatically approve a release.

- If there is not enough evidence to make a confident recommendation, the system should clearly state what needs further review.

Testing Requirement

You are responsible for designing a suitable testing strategy.

Your tests should demonstrate that the system can respond appropriately when the release situation changes.

Testing should be detailed enough to show that your system is using the release rules and evidence correctly rather than always producing the same recommendation.

You should test situations involving changes such as:

- different test results;
- different defect severity levels;
- new unresolved defects;
- removed defects;
- missing QA information;
- different business impacts;
- changes to available workarounds; and
- conflicting release information.

Keep evidence of your testing and any improvements you make.

Testing Record

You may use this format or create your own.

Test	What You Tested	Expected Result	Actual Result	Issue Found	Improvement Made
1					
2					
3					
4					

Final Project Submission

Your final submission should demonstrate your complete problem-solving process.

1. Problem Understanding

Explain the release-readiness problem faced by NextWave Software.

2. Existing Process

Explain how Development, QA, and DevOps teams would normally review a release without your system.

3. AI Opportunity

Explain where AI can support the release-review process and where normal automation or human judgment is more suitable.

4. Solution Design

Explain the system you designed and why you chose that approach.

5. Multi-Agent Design

Show the agents you created, their responsibilities, and how they work together.

6. Workflow

Show how release data moves through your system until the final assessment is produced.

7. Working System

Demonstrate the actual Multi-Agent System.

8. Evidence Use

Show how your system uses changes, test results, defects, business impact, workarounds, and release rules.

9. Testing and Improvement

Show how you tested the system, what problems you found, and what you changed.

10. Final Release Assessment

Show at least one complete release-readiness report produced by your system.

11. Business Impact

Explain how your solution could help Development, QA, DevOps, and technical leads make release reviews faster or more consistent.

12. Limitations

Explain what the AI may misunderstand, what information it cannot verify, and where incorrect conclusions could occur.

13. Human Oversight

Explain which decisions must remain with the Technical Lead and why.

Final Presentation

Your presentation should clearly communicate the following.

The Problem

What release-readiness problem are you solving?

Your Approach

How did you analyse the problem and decide what your Multi-Agent System needed to do?

Your Multi-Agent System

What agents did you create, what does each one do, and why did you structure the system that way?

Live Demo

Run the system using the provided release data.

Show how the system uses the available information to prepare a release-readiness assessment.

Testing

Demonstrate how the system behaves when test results, defects, severity, or release information changes.

Impact and Limitations

Explain the value of the solution, the limitations of AI-based release assessment, and why final release authority must remain with the Technical Lead.

Your Goal

The goal is not to build a system that automatically releases software.

The goal is to build a system that helps technical teams review release evidence more effectively.

You are expected to understand the business and technical problem, design the Multi-Agent System, decide how release evidence should be evaluated, choose appropriate tools, test your solution, and improve it based on what you discover.

A strong project should demonstrate technical reasoning, careful use of evidence, responsible AI design, and clear human oversight.